

QRM II Graded Assignment (7)

Period 1, 2026

Peter Ac, Jakob Vahtera Persson, Shabir Adel, David Ruivo Craveiro

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0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th**. You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

1 Week 1

1. Find the dataset “movies3.tsv” on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject? [4 points]

```
movies <- read.delim("movies3.tsv")
dim(movies)
```

```
## [1] 707 19
```

Your Answer:

Based on the (dim) output, the dataset has 707 observations and 19 variables. Each row represents a specific movie, meaning that there are 707 subjects in total. A subject in this specific context is simply the movie itself.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); interval; (iv) ratio. If present, name one example of such a variable present in your data set. [4 points]

```
#str(movies)
```

Your Answer:

- (i) Genre or original_language, since these represent categories without any numerical order.
- (ii) There are no strictly ordinal variables present in this dataset.
- (iii) Release_year, because the difference between years is mathematically meaningful, however there is no “true zero”, as the year 0 does not reflect absence of time.
- (iv) Budget, revenue, and runtime. Because these are continuous numerical values where a true zero suggest zero dollars or zero minutes.

3. A movie studio wants to know which types of movies are best at generating profit. The measure they use for this is the profitability ratio, which is defined as

$$PR = \frac{Revenue}{Budget}.$$

Perform the following steps to provide the movie studio with an analysis which corresponds to their request:

- Create the variable PR as the revenue of a movie divided by its budget. Report its mean, median, maximum, and minimum. **[2 points]**
- As you can probably see from your answer in a.), some movies have a PR of “Inf”. What does this mean? Why are some movies assigned this value? **[2 points]**
- As you can probably see from your answer in a.), some movies have a PR of “NA”. What does this mean? Why are some movies assigned this value? **[2 points]**
- For the movies that have a PR of “Inf”, set its PR to “NA”. After doing this, report the mean, median, maximum, and minimum of PR. How do you interpret the mean? **[2 points]**
- What does a PR of at least one mean? What is the fraction of movies that have a PR of at least one? **[2 points]**
- Create a boxplot of the variable PR. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Describe the shape of the boxplot. Give Q_1 , Q_2 , Q_3 and Q_4 . What does this tell you about the nature of making money in the movies industry? **[2 points]**

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Step a

```
movies$PR <- movies$revenue / movies$budget
mean(movies$PR, na.rm = TRUE)
```

```
## [1] Inf
```

```
median(movies$PR, na.rm = TRUE)
```

```
## [1] 1.665002
```

```
max(movies$PR, na.rm = TRUE)
```

```
## [1] Inf
```

```
min(movies$PR, na.rm = TRUE)
```

```
## [1] 0
```

Your Answer:

The mean PR is Inf (infinity) and the median is 1.665002. The maximum PR is also Inf, and the minimum PR is 0.

Step b

```
#movies[is.infinite(movies$PR), c("title", "budget", "revenue", "PR")]
```

Your Answer:

An “Inf” value means that the formula tried to divide a positive number by zero. By looking at the data, this happens as some movies have a recorded budget of 0, but a positive revenue. It is important to note that this does not mean that there was zero production cost, it just means that the budget data is missing and was incorrectly entered into the dataset as a zero.

Step c

```
#movies[is.na(movies$PR), c("title", "budget", "revenue", "PR")]
```

Your Answer:

“NA” stands for Not Available. This outcome happens when R tries to divide zero by zero, which yields NaN (a type of NA). This correctly reflects these specific movies where both budget and revenue were recorded as a zero.

Step d

```
movies$PR[is.infinite(movies$PR)] <- NA  
mean(movies$PR, na.rm = TRUE)
```

```
## [1] 1813.041
```

```
median(movies$PR, na.rm = TRUE)
```

```
## [1] 1.55807
```

```
max(movies$PR, na.rm = TRUE)
```

```
## [1] 1e+06
```

```
min(movies$PR, na.rm = TRUE)
```

```
## [1] 0
```

Your Answer:

After replacing the infinite values with NAs, the new mean PR is 1813.041, the median is 1.55807, the max is 1,000,000, and the min is 0.

The mean of 1,813 suggests that the average movie makes over 1,800 times its budget. However, by observing how far the mean is pulled away from the median, we can conclude that the mean is massively distorted by some extreme outliers and is not a fully reliable measure of a movie’s performance.

Step e

```
mean(movies$PR >= 1, na.rm = TRUE)
```

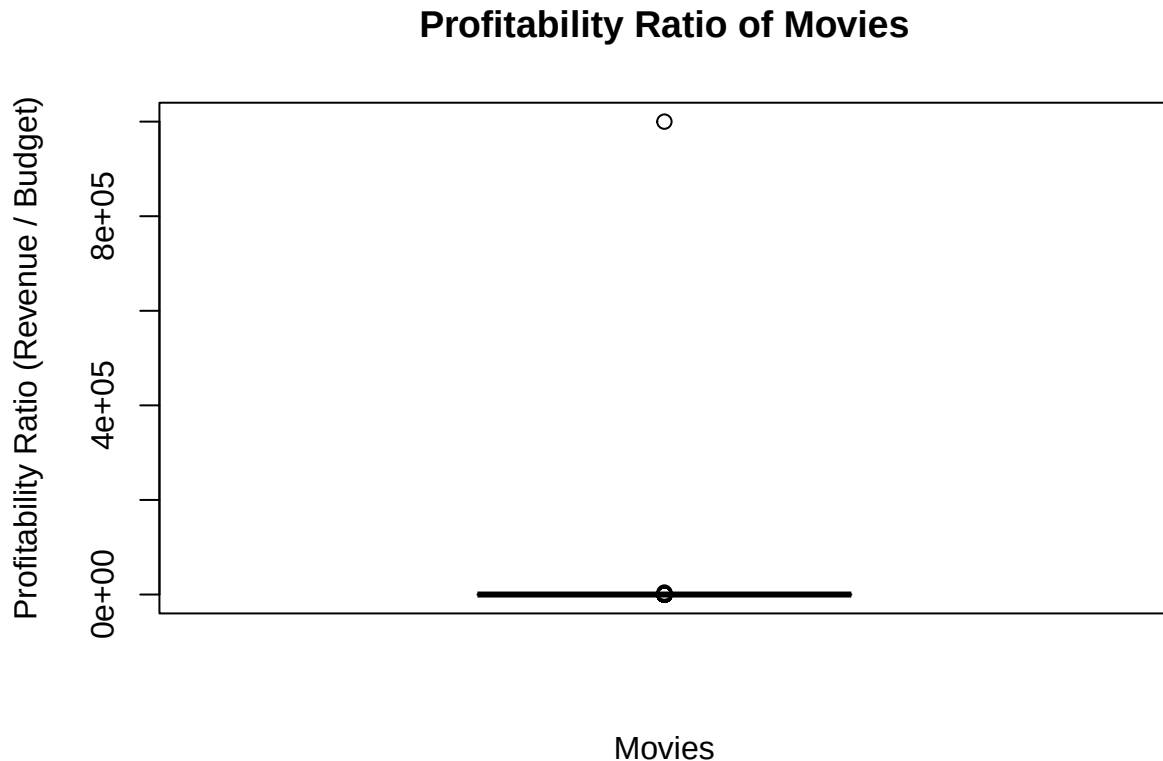
```
## [1] 0.6036036
```

Your Answer:

A PR of 1 or higher implies that the movie reached break-even or turned a profit. In this sample, approximately 60.36% of the movies achieved this.

Step f

```
boxplot(movies$PR,
        main = "Profitability Ratio of Movies",
        xlab = "Movies",
        ylab = "Profitability Ratio (Revenue / Budget)")
```



```
quantile(movies$PR,
        probs = c(0.25, 0.50, 0.75, 1),
        na.rm = TRUE)
```

```
##          25%          50%          75%          100%
## 3.344855e-01 1.558070e+00 3.370324e+00 1.000000e+06
```

Your Answer:

The boxplot is heavily right-skewed. Essentially all is compressed into a flat line at the bottom near zero, but there is at least one extreme outlier pulling the plot scale up to 1,000,000.

Q1 (25%): 0.33 Q2 (Median): 1.56 Q3 (75%): 3.37 Q4 (Max): 1,000,000

This data reveals that making money in the movie industry is tough. The vast majority of movies make very modest returns, which is depicted in the third quantile, but a tiny fraction of movies becomes immensely successful and thus skew the industry's numbers.

2 Week 2

1 Is your dataset movies3.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? [4 points]

Answer: The dataset is a sample, not the full population. The variable index runs from 2 to 4798 with gaps, but we only have 707 movies. So our movies were drawn from a list of about 4,800 movies (roughly 15% of it). Variables such as vote_average, vote_count, popularity and keywords come from The Movie Database (TMDB). The population is therefore most likely the TMDB 5000 movie database, limited to films released between 1990 and 2016 (our data contain no other years). More broadly, it is all commercially released feature films in that period, mostly English-language.

2 Let M be the event that a movie director is male. Let R be the runtime of the movie.

- Using your data, calculate $P(M)$. What is $P(M')$? Calculate $P(M')$ as well. [1 point]
- Assume R is normally distributed with $\mu = 107$ and $\sigma = 22$, calculate $P(90 \leq R \leq 120)$. [1 point]
- Calculate $P(90 \leq R \leq 120)$ empirically using your data. How accurate were the assumptions about the distribution of R in 2b? Which of the assumptions do you consider the most suspect? Explain. [2 points]
- Using your data, calculate $P(120 \leq R|M')$ [2 points]

step a

```
#WRITE YOUR CODE HERE
table(movies$director_gender, useNA = "ifany")
```

```
##
## female    male    <NA>
##      65     588     54
```

```
P_M <- mean(movies$director_gender == "male", na.rm = TRUE)
P_M
```

```
## [1] 0.9004594
```

```
P_Mc <- 1 - P_M
P_Mc
```

```
## [1] 0.09954058
```

Your Answer: Answer: 653 movies have a known director gender: 588 male and 65 female. The 54 unknowns are left out. $P(M) = 588/653 = 0.9005$. M' is the complement of M : the director is not male, which in this data means female. By the complement rule, $P(M') = 1 - P(M) = 65/653 = 0.0995$.

step b

```
#WRITE YOUR CODE HERE
pnorm(120, mean = 107, sd = 22) - pnorm(90, mean = 107, sd = 22)
```

```
## [1] 0.5028674
```

Your Answer: If $R \sim N(107, 22^2)$, then $P(90 \leq R \leq 120) = P(-0.77 \leq Z \leq 0.59) = \Phi(0.59) - \Phi(-0.77) = 0.5029$. So about 50.3% of movies would run between 90 and 120 minutes. Write your formulated response here.

step c

```
#WRITE YOUR CODE HERE
movies$runtime[movies$runtime == 0] <- NA
mean(movies$runtime >= 90 & movies$runtime <= 120, na.rm = TRUE)
```

```
## [1] 0.6857963
```

```
m <- mean(movies$runtime, na.rm = TRUE)
s <- sd(movies$runtime, na.rm = TRUE)
m; s
```

```
## [1] 104.9225
```

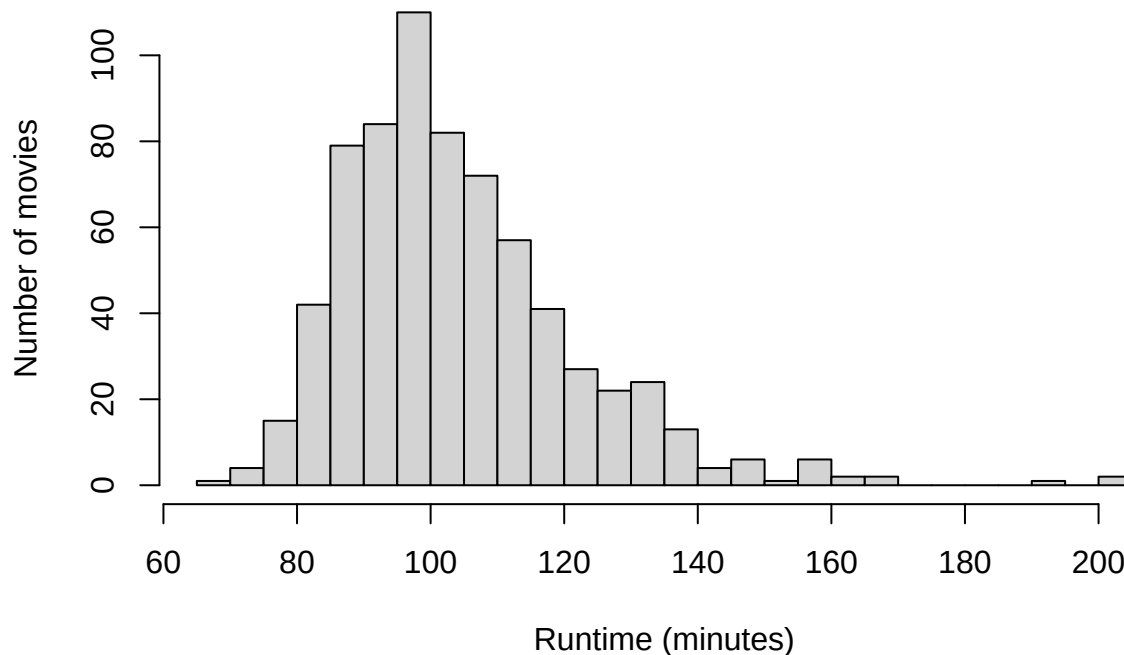
```
## [1] 17.67335
```

```
pnorm(120, m, s) - pnorm(90, m, s)
```

```
## [1] 0.6039669
```

```
hist(movies$runtime, breaks = 30, main = "Distribution of movie runtime",
      xlab = "Runtime (minutes)", ylab = "Number of movies")
```

Distribution of movie runtime



Your Answer: 478 of the 697 movies with a known runtime last between 90 and 120 minutes, so $P(90 \leq R \leq 120) = 0.686$. We treated the 10 movies with runtime 0 as missing. The normal model from 2b (0.503) underestimates this by about 18 percentage points, so its assumptions were not accurate. -The assumed mean is roughly right: the sample mean is 104.9, against 107. -The assumed standard deviation is too large: the sample sd is 17.7, against 22. -The most suspect assumption is normality. Even with the sample's own mean and sd, a normal curve gives only 0.604.

The histogram shows why. Runtimes are right-skewed (mean 104.9 > median 101): only 1% of movies are shorter than 78 minutes, but a long tail runs up to 202 minutes. Most movies are packed between 90 and 120 minutes. A symmetric bell curve puts too much probability in the tails so it predicts 22% of movies under 90 minutes and 28% over 120, while the real shares are 16% and 16%.

step d

```
#WRITE YOUR CODE HERE
female_dir <- subset(movies, director_gender == "female")
sum(!is.na(female_dir$runtime)) # female-directed movies with known runtime
```

```
## [1] 64
```

```
sum(female_dir$runtime >= 120, na.rm = TRUE) # of those have a runtime of 120+ minutes
```

```
## [1] 9
```

```
mean(female_dir$runtime >= 120, na.rm = TRUE)
```

```
## [1] 0.140625
```

Your Answer: $P(R = 120 | M') = P(R = 120 \cap M') / P(M')$. Conditioning on M' restricts the sample space to movies with a female director. 64 of them have a known runtime, and 9 of those last 120 minutes or more. So $P(R = 120 | M') = 9/64 = 0.141$.

3 For this question, you will again use the profitability ratio PR. Make sure that PR is defined the same way as in question 3d of week 1. For this question, you will assume that your data set is the full population.

- What is the mean of the variable PR in your data set? **[2 points]**
- Create a new data set, called `movies_sample`. Make sure that it is a random sample of your data set of 25 movies. What is the mean of PR in this random sample? How does it compare to the mean of PR in 3a? **[2 points]**
- In a for loop, create 100 different samples of 25 movies, as in b, and estimate the mean within each sample. Save the mean of each sample in a vector called `sample_means`. So the first position of the vector would have the mean of the first sample, the second position the mean of the second sample, etc. Print this vector. **[2 points]**
- Summarize and make a histogram of `sample_means`. What is the mean, standard deviation and shape of its distribution? **[2 points]**
- Using the formulas of the *Central Limit Theorem*, give the distribution of the sample mean of PR in a sample of 25 movies from you population (including its expected value and standard deviation). How do these values compare to your answers at d.)? **[2 points]**

step a

```
#WRITE YOUR CODE HERE
movies$PR <- movies$revenue / movies$budget
movies$PR[is.infinite(movies$PR)] <- NA
movies_pop <- movies[!is.na(movies$PR), ] # population: the movies that have a PR
nrow(movies_pop)
```

```
## [1] 555
```

```
mean(movies_pop$PR)
```

```
## [1] 1813.041
```

```
mean(movies_pop$PR[movies_pop$title != "Nurse 3-D"]) # mean without the $10-budget typo
```

```
## [1] 11.25908
```

Your Answer: Treating the data as the population ($N = 555$ movies with a defined PR), the population mean is $\bar{x} = 1,813.04$. This mean comes almost entirely from Nurse 3-D, whose budget is recorded as 10\$ instead of 10 million dollars, giving $PR = 1,000,000$. Without that movie the mean would be 11.26.

step b

```
#WRITE YOUR CODE HERE
set.seed(123)
movies_sample <- movies_pop[sample(nrow(movies_pop), 25), ]
mean(movies_sample$PR)
```

```
## [1] 40003.65
```

```
movies_sample[movies_sample$PR > 100, c("title", "budget", "revenue", "PR")]
```

```
##           title budget revenue    PR
## 300 Nurse 3-D      10    1e+07 1e+06
```

Your Answer: The mean PR in our random sample of 25 movies is 40,003.65, about 22 times the population mean of 1,813.04. The reason is that Nurse 3-D is in this sample. On its own it adds $1,000,000/25 = 40,000$ to the sample mean, while the other 24 movies average only 3.80. So the sample mean behaves very badly here. Any sample of 25 either contains Nurse 3-D, giving a mean of about 40,000, or it does not, giving a mean below about 200 and usually around 3. A sample mean can never land near 1,813.

step c

```
#WRITE YOUR CODE HERE
set.seed(456)
sample_means <- numeric(100)
for (i in 1:100) {
  one_sample <- movies_pop[sample(nrow(movies_pop), 25), ]
  sample_means[i] <- mean(one_sample$PR)
}
```

Your Answer:

sample_means holds the means of 100 random samples of 25 movies. Most values lie between 1.25 and about 7. Five samples are around 19 and eight are around 168. Nine samples are around 40,000: samples 1, 9, 34, 44, 52, 56, 57, 59 and 79.

step d

```
#WRITE YOUR CODE HERE
summary(sample_means)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##   1.254    2.443    3.180   3617.336    6.400  40023.033
```

```
mean(sample_means)
```

```
## [1] 3617.336
```

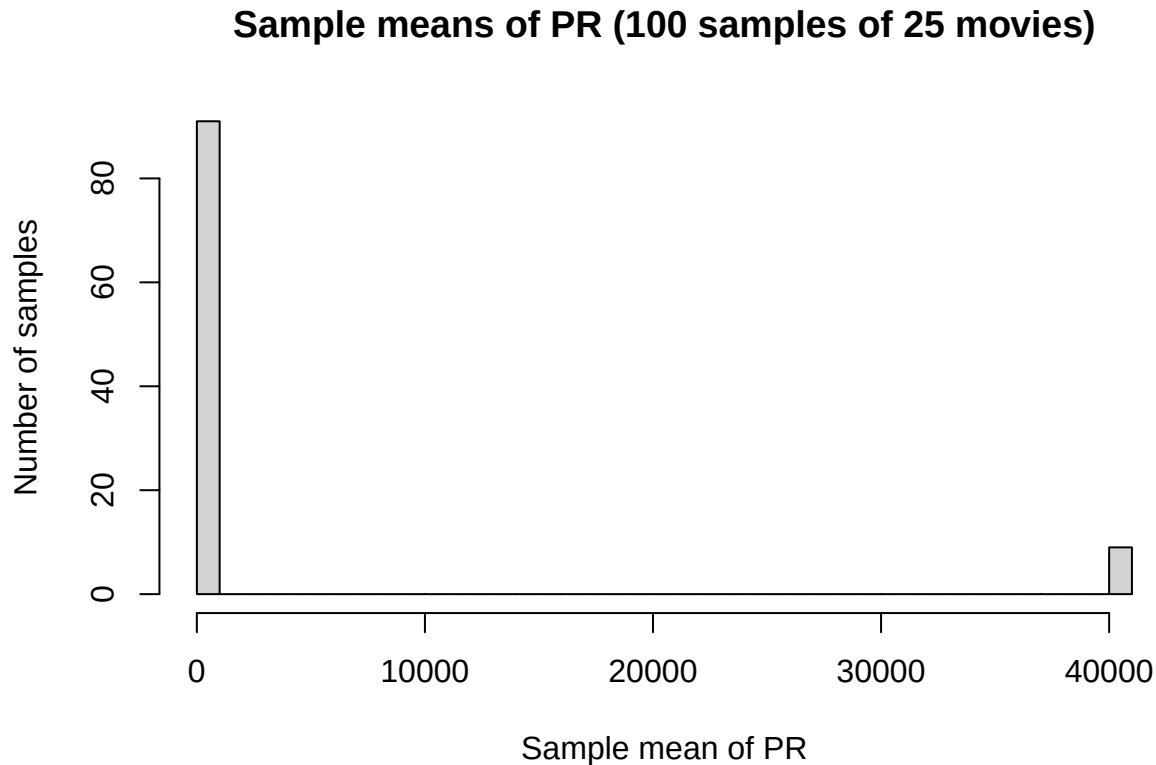
```
sd(sample_means)
```

```
## [1] 11501.31
```

```
sum(sample_means > 1000)  # samples that contain Nurse 3-D
```

```
## [1] 9
```

```
hist(sample_means, breaks = 50, main = "Sample means of PR (100 samples of 25 movies)",  
      xlab = "Sample mean of PR", ylab = "Number of samples")
```



Your Answer: Mean and spread: the mean of the sample means is 3,617.34 and their standard deviation is 11,501.31.

Median and quartiles: the median is only 3.18 (min 1.25, Q1 2.44, Q3 6.40, max 40,023.03).

Shape: extremely right-skewed and not at all normal. In the histogram, 91 samples form one bar near 0, and 9 samples form a separate bar near 40,000, with nothing in between. Those 9 are exactly the samples that contain Nurse 3-D. A mean more than 1,000 times the median confirms the extreme skew.

step e

```
#WRITE YOUR CODE HERE  
mu    <- mean(movies_pop$PR)  
sigma <- sd(movies_pop$PR)  
se    <- sigma / sqrt(25)  
mu; sigma; se
```

```
## [1] 1813.041
```

```
## [1] 42447.52
```

```
## [1] 8489.505
```

```
pnorm(0, mean = mu, sd = se) # CLT probability of a negative sample mean
```

```
## [1] 0.4154441
```

Your Answer: By the Central Limit Theorem, $X \sim N(\mu, \sigma^2/n)$. Here $E(X) = \mu = 1,813.04$ and $SD(X) = \sigma/\sqrt{n} = 42,447.52/\sqrt{25} = 8,489.51$, so $X \sim N(1,813.04; 8,489.51^2)$.

Compared with d:

Mean: the simulation gives 3,617 instead of 1,813. The mean is off because 9 of our 100 samples contained Nurse 3-D, while on average only 4.5 would. With just 100 samples, one rare movie decides the result.

Standard deviation: the simulation gives 11,501 instead of 8,490.

Shape: the simulated distribution is not normal at all. The CLT only works when n is large enough. The course's rule of thumb ($n \geq 30$ normally) assumes no extreme outliers. Our population is extremely skewed with one extreme value, so $n = 25$ is far too small. The normal approximation even gives a 41.5% chance of a negative sample mean, which is impossible because $PR \leq 0$.

3 Week 3

1. For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.
 - a. Create a new data set that only includes movies that are of the genre "Thriller". For these thriller movies, give a 99 percent confidence interval for the variable `vote_average`. [2 points]
 - b. According to IMDB, the average movie is rated with a 6.5. Using your data, test for the null hypothesis that thriller movies are at least as good as the average movie, against the alternative hypothesis that thriller movies are rated worse than the average movie. Test this hypothesis using an appropriate five-step procedure. What is your p-value and what do you conclude? [2 points]

step a

```
thriller_movies <- subset(movies, genre == "Thriller")
mean(thriller_movies$vote_average, na.rm = TRUE)
```

```
## [1] 5.890722
```

```
sd(thriller_movies$vote_average, na.rm = TRUE)
```

```
## [1] 0.8270281
```

```
sum(!is.na(thriller_movies$vote_average))
```

```
## [1] 97
```

```
t.test(thriller_movies$vote_average, conf.level = 0.99)
```

```
##
## One Sample t-test
##
## data: thriller_movies$vote_average
## t = 70.151, df = 96, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 99 percent confidence interval:
## 5.670042 6.111401
## sample estimates:
## mean of x
## 5.890722
```

Your Answer:

The 99% confidence interval is [5.670042, 6.111401].

step b

```
t.test(thriller_movies$vote_average,
       mu = 6.5,
       alternative = "less",
       conf.level = 0.95)
```

```
##
## One Sample t-test
##
## data: thriller_movies$vote_average
## t = -7.2557, df = 96, p-value = 5.152e-11
## alternative hypothesis: true mean is less than 6.5
## 95 percent confidence interval:
##      -Inf 6.030189
## sample estimates:
## mean of x
## 5.890722
```

```
#Because alternative hypothesis is less than 6.5, this is a left-tailed test.
#Significance level is 0.05.
```

Your Answer:

The P-value is 5.152e-11, indicating that the value is practically zero and does hence fall below the alpha threshold. As a result, we reject the null hypothesis.

There is sufficient evidence to conclude that the true mean of average vote of thriller movies, is below 6.5, supported by the low P-value.

2

- a. How many movies in your sample had a female lead actor? How many had a male lead actor? [2 points]

- b. Test the null hypothesis that for any given movie the chance that the lead actor is male or female is equal. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Use an appropriate R function to test your hypothesis, and state your conclusion. Also give the p-value that corresponds to your test statistic. **[2 points]**
- c. What is the variance of the profitability ratio (PR) for movies with a male lead actor? What is the variance of profitability ratio for movies with a female lead actor? Make sure that PR is defined in the same way as in week 1. **[2 points]**
- d. Assume that the variance of PR for movies with a male lead actor in your data is the same as the variance for movies with a male lead actor in your population (that is, there is no sampling uncertainty in this estimate). Now, set up a hypothesis test to test for the null hypothesis that the PR for movies with a female lead is equal to the variance for movies with a male lead actor. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Calculate your test statistic and tell how your test statistic is distributed, and state your conclusion. Also give the p-value that corresponds to your test statistic. **[2 points]**

step a

```
table(movies$first_actor_gender, useNA = "ifany")
```

```
##
## female    male    <NA>
##      182     495      30
```

Your Answer:

There were 182 films with a female as the lead actor, and 495 starring a male as the leading actor.

step b

```
num_female <- sum(movies$first_actor_gender == "female", na.rm = TRUE)
num_male <- sum(movies$first_actor_gender == "male", na.rm = TRUE)
num_total <- num_female + num_male
binom.test(x = num_female, n = num_total, p = 0.5, alternative = "two.sided")
```

```
##
## Exact binomial test
##
## data: num_female and num_total
## number of successes = 182, number of trials = 677, p-value < 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.2357667 0.3039290
## sample estimates:
## probability of success
##                0.2688331
```

Your Answer:

$H_0: p = 0.5 \rightarrow$ probability that the lead actor is male vs female is equal $H_1: p \neq 0.5$ $\alpha = 0.05$ significance level Out of 677 movies only 182 lead actors are female. The binomial test statistic of 182 successes in 677 trials yields a p-value of $2.2e-16$, this value is below the significance level of 0.05 and therefore the null hypothesis is rejected. In conclusion there is enough statistical evidence that the chance of a movies lead actor being female vs male is not equal.

step c

```

movies$PR <- movies$revenue / movies$budget
movies$PR[is.infinite(movies$PR)] <- NA
var_male <- var(movies$PR[movies$first_actor_gender == "male"], na.rm = TRUE)
var_female <- var(movies$PR[movies$first_actor_gender == "female"], na.rm = TRUE)
var_male

```

```
## [1] 43030.08
```

```
var_female
```

```
## [1] 6992928750
```

Your Answer:

The sample variance of the profitability ratio for movies with a male lead actor is 43,030.08 whilst the sample variance of the profitability for movies with a female lead actor is 6,992,928,750.

step d

```

sigma0_sq <- var(movies$PR[movies$first_actor_gender == "male"], na.rm = TRUE)
pr_female <- na.omit(movies$PR[movies$first_actor_gender == "female"])
n_f <- length(pr_female)
s_sq_f <- var(pr_female)
test_stat <- (n_f - 1) * s_sq_f / sigma0_sq
df <- n_f - 1
p_val <- 2 * (1 - pchisq(test_stat, df = df))

```

Your Answer:

We set up a two-sided hypothesis test where the null hypothesis is that the variance of the profitability ratio for female-lead equals the variance for male-lead movies, and the alternative hypothesis is that they are not equal. We use a significance level $\alpha = 0.05$.

By using our data and the male variance as the population value, the test statistic follows a chi-squared distribution with $n-1$ degrees of freedom. The test then yields a p-value of basically zero, which is way below 0.05. Hence, we reject the null hypothesis. There is thus clear evidence that the variance for female-lead movies is different than that of male-lead movies. It is much larger due to the extreme values in that subset.

3

- Create a new variable called `genre2`. Make sure that the three most popular genres remain the same, but that all the other genres get categorized into one genre called "Other". Report a frequency table of this new variable `genre2`. **[2 points]**
- Create a scatter plot with each type of `genre2` on the x axis and the mean of PR within that genre on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Which movie genre has the highest PR? Which has the lowest? **[2 points]**
- Recreate the scatter plot, but now add bars around each point, indicating the 95% confidence interval. **[2 points]**
- Now, create a new data frame which is a random subsample of $n=100$ of your full data frame. Recreate the plot in c.) How do the confidence intervals compare to the confidence intervals you plotted in c.)? **[2 points]**

step a

```
top3_genres <- names(sort(table(movies$genre), decreasing = TRUE)[1:3])
movies$genre2 <- ifelse(movies$genre %in% top3_genres, movies$genre, "Other")
movies$genre2 <- factor(movies$genre2)
table(movies$genre2)
```

```
##
##   Comedy   Drama   Other Thriller
##      176     332     102      97
```

Your Answer:

The most frequent genres are Drama, Comedy and Thriller with 322, 176 and 97 occurrences in the data set respectively

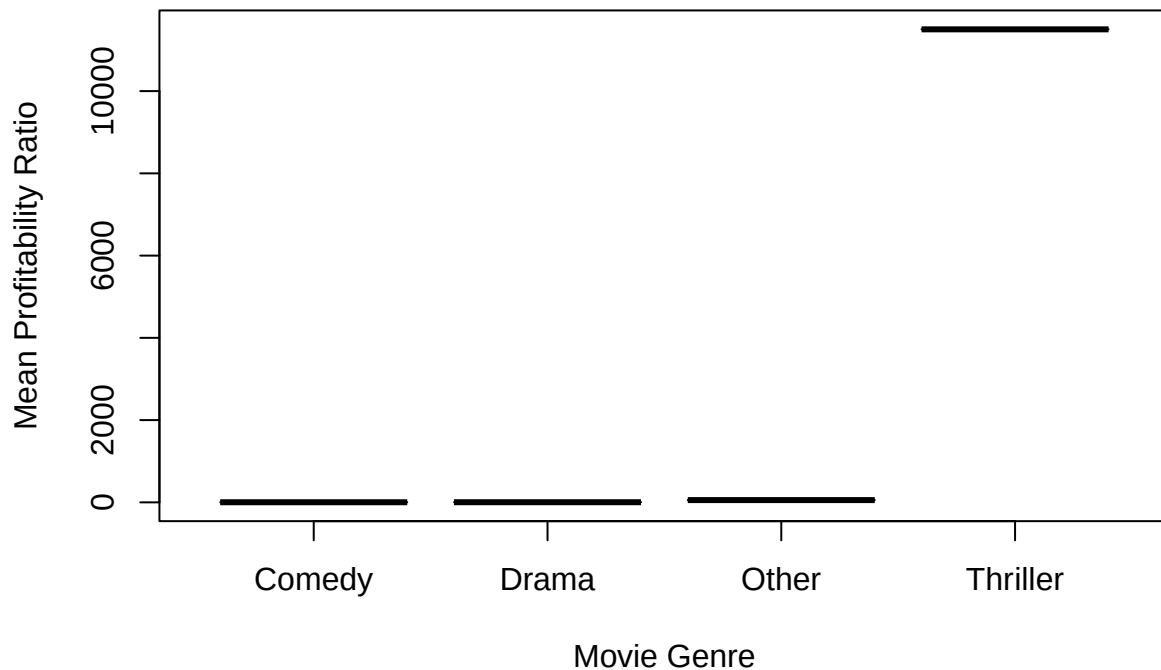
step b

```
genre_means <- aggregate(PR ~ genre2, data = movies, FUN = mean, na.rm = TRUE)
genre_means
```

```
##      genre2      PR
## 1  Comedy  2.684404
## 2   Drama  3.318001
## 3   Other 54.872781
## 4 Thriller 11501.683848
```

```
plot(genre_means$genre2, genre_means$PR, main = "Mean Profitability Ratio (PR) by Genre",
     ↪ xlab = "Movie Genre", ylab = "Mean Profitability Ratio", pch = 19, col = "blue", cex
     ↪ = 1.5)
```

Mean Profitability Ratio (PR) by Genre



Your Answer:

Thriller has by far the highest PR which is skewed by the outlier Nurse 3-D (1,000,000 PR)

step c

```
genre_summary <- do.call(data.frame, aggregate(PR ~ genre2, data = movies, function(x) {
  x <- na.omit(x)
  n <- length(x)
  m <- mean(x)
  se <- sd(x) / sqrt(n)
  ci_low <- m - qt(0.975, df = n - 1) * se
  ci_high <- m + qt(0.975, df = n - 1) * se
  c(mean = m, ci_low = ci_low, ci_high = ci_high)
}))
colnames(genre_summary) <- c("genre2", "mean", "ci_low", "ci_high")
genre_summary
```

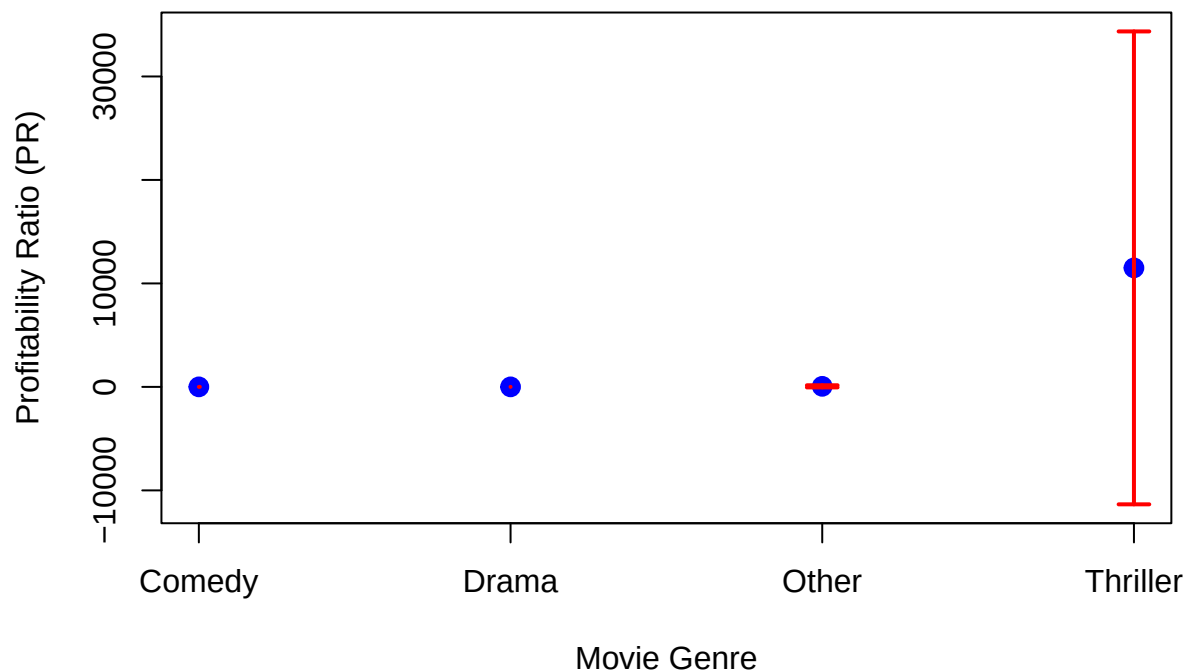
```
##      genre2      mean      ci_low      ci_high
## 1  Comedy    2.684404    2.063409    3.305399
## 2   Drama    3.318001    2.132932    4.503069
## 3   Other   54.872781   -47.892062   157.637624
## 4 Thriller 11501.683848 -11347.964905 34351.332601
```

```
y_lims <- range(c(genre_summary$ci_low, genre_summary$ci_high))
plot(1:nrow(genre_summary), genre_summary$mean, xaxt = "n", ylim = y_lims, pch = 19, col
  ↪ = "blue", cex = 1.3, main = "Mean PR with 95% Confidence Intervals by Genre", xlab =
  ↪ "Movie Genre", ylab = "Profitability Ratio (PR)")
```

```
axis(1, at = 1:nrow(genre_summary), labels = genre_summary$genre2)
arrows(x0 = 1:nrow(genre_summary), y0 = genre_summary$ci_low,
       x1 = 1:nrow(genre_summary), y1 = genre_summary$ci_high,
       angle = 90, code = 3, length = 0.08, col = "red", lwd = 2)
```

```
## Warning in arrows(x0 = 1:nrow(genre_summary), y0 = genre_summary$ci_low, :
## zero-length arrow is of indeterminate angle and so skipped
## Warning in arrows(x0 = 1:nrow(genre_summary), y0 = genre_summary$ci_low, :
## zero-length arrow is of indeterminate angle and so skipped
```

Mean PR with 95% Confidence Intervals by Genre



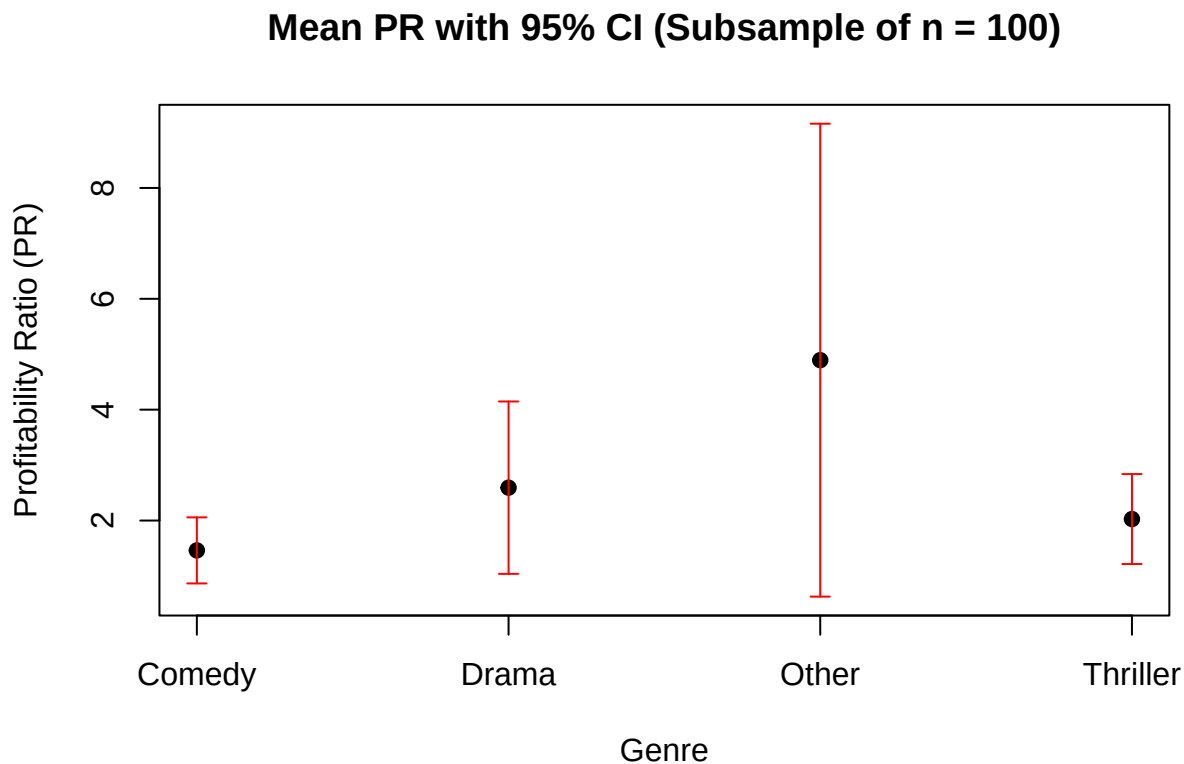
Your Answer:

The categories Other and Thriller have significant standard errors due to the previously mentioned outliers such as Nurse 3-D. This movie on its own is largely responsible for the wide confidence interval for the Thriller category.

step d

```
set.seed(123)
movies_sub <- movies[sample(nrow(movies), 100), ]
mean_sub <- tapply(movies_sub$PR, movies_sub$genre2, mean, na.rm = TRUE)
sds_sub <- tapply(movies_sub$PR, movies_sub$genre2, sd, na.rm = TRUE)
ns_sub <- tapply(movies_sub$PR, movies_sub$genre2, function(x) sum(!is.na(x)))
se_sub <- sds_sub / sqrt(ns_sub)
ci_low_sub <- mean_sub - 1.96 * se_sub
ci_high_sub <- mean_sub + 1.96 * se_sub
```

```
plot(1:4, mean_sub, xaxt = "n", pch = 19, ylim = range(c(ci_low_sub, ci_high_sub), na.rm
  → = TRUE), main = "Mean PR with 95% CI (Subsample of n = 100)", xlab = "Genre", ylab =
  → "Profitability Ratio (PR)")
axis(1, at = 1:4, labels = names(mean_sub))
arrows(1:4, ci_low_sub, 1:4, ci_high_sub, angle = 90, code = 3, length = 0.05, col =
  → "red")
```



Your Answer:

Compared to the plot in part c, the confidence intervals in this n=100 sub-sample are visibly more wide and spread out.

This makes sense due to the standard error being calculated by dividing the standard deviation by the square root of n. By reducing our sample size down to 100, we decrease the denominator, which then causes the standard errors to increase and makes our 95% confidence intervals wider. In addition, the point estimates does jump around a lot more because a smaller random sample is more sensitive to whichever specific movies get picked.

4 Week 4

1. There is an argument going on in the movie studio. *Bob* claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. *Chantal* disagrees. She tells Bob that “The profitability ratio of the movie will remain similar, no matter how much the budget is’.

- a. Set up a regression model to test Chantal's claim, and estimate it. That is, estimate:

$$PR_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i.$$

Print a summary of your estimated model. [2 points]

- b. What is the estimated value of β_1 how do you interpret it? [2 points]
 c. Test for the null hypothesis that $\beta_1 = 0$. Report the p-value and state your conclusion. [2 points]
 d. Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? [2 points]
 e. Make a histogram of the residuals of your model. Based on the histogram, do you have any reason to doubt the validity of your model? Explain. [2 points]
 f. Estimate hatvalues for each observation that you used to estimate your model. Do any hatvalues exceed the critical value? Based on this analysis, do you have any reason to doubt the validity of your model? [2 points]
 g. Based on your analysis in f.), estimate your model again, but remove any outliers. Print a summary of your model and again test for the null hypothesis that $\beta_1 = 0$. What do you conclude? Based on this new model, would your advice to the movie studio in question d. change? [2 points]

step a

```
#WRITE YOUR CODE HERE
model1 <- lm(PR ~ budget, data = movies)
summary(model1)

##
## Call:
## lm(formula = PR ~ budget, data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3235   -2866   -2312   -1314   996765
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.235e+03  2.408e+03   1.343   0.180
## budget      -3.689e-05  4.143e-05  -0.890   0.374
##
## Residual standard error: 42460 on 553 degrees of freedom
## (152 observations deleted due to missingness)
## Multiple R-squared:  0.001431,    Adjusted R-squared:  -0.0003744
## F-statistic: 0.7927 on 1 and 553 DF,  p-value: 0.3737
```

Your Answer:

The estimated model is $PR = 3,234.9 - 0.0000369 \times \text{Budget}$, with $n = 555$; the 152 movies without a PR are dropped automatically. $R^2 = 0.0014$, so budget explains only 0.14% of the variation in PR.

step b

```
#WRITE YOUR CODE HERE
coef(model1)[ "budget" ]

##           budget
## -3.688924e-05
```

```
coef(model1)["budget"] * 1000000 # effect of 1 million dollars extra budget
```

```
##      budget  
## -36.88924
```

Your Answer: $\hat{1} = -0.0000369$ ($-3.689\text{e-}05$). Each extra dollar of budget goes with an expected PR that is 0.0000369 lower. Per extra \$1 million of budget, the expected PR is 36.9 lower. That sounds large, but the estimate is driven by the outlier Nurse 3-D and is not significant.

step c

```
#WRITE YOUR CODE HERE  
summary(model1)$coefficients["budget", ]
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)  
## -3.688924e-05  4.143360e-05 -8.903220e-01  3.736802e-01
```

Your Answer:

$H_0: \beta_1 = 0$ (PR does not depend on budget, which is Chantal's claim). $H_1: \beta_1 \neq 0$. $\alpha = 0.05$. The test statistic is $t = \hat{\beta}_1 / \text{SE}(\hat{\beta}_1) = -3.689\text{e-}05 / 4.143\text{e-}05 = -0.89$. Under H_0 , t follows a t -distribution with $n - 2 = 553$ degrees of freedom. $p\text{-value} = 0.374 > 0.05$, and $|t| = 0.89 < 1.96$. We do not reject H_0 . There is no significant evidence that budget is related to PR.

step d

```
#WRITE YOUR CODE HERE  
# no new code needed -- the answer follows from a) to c) !
```

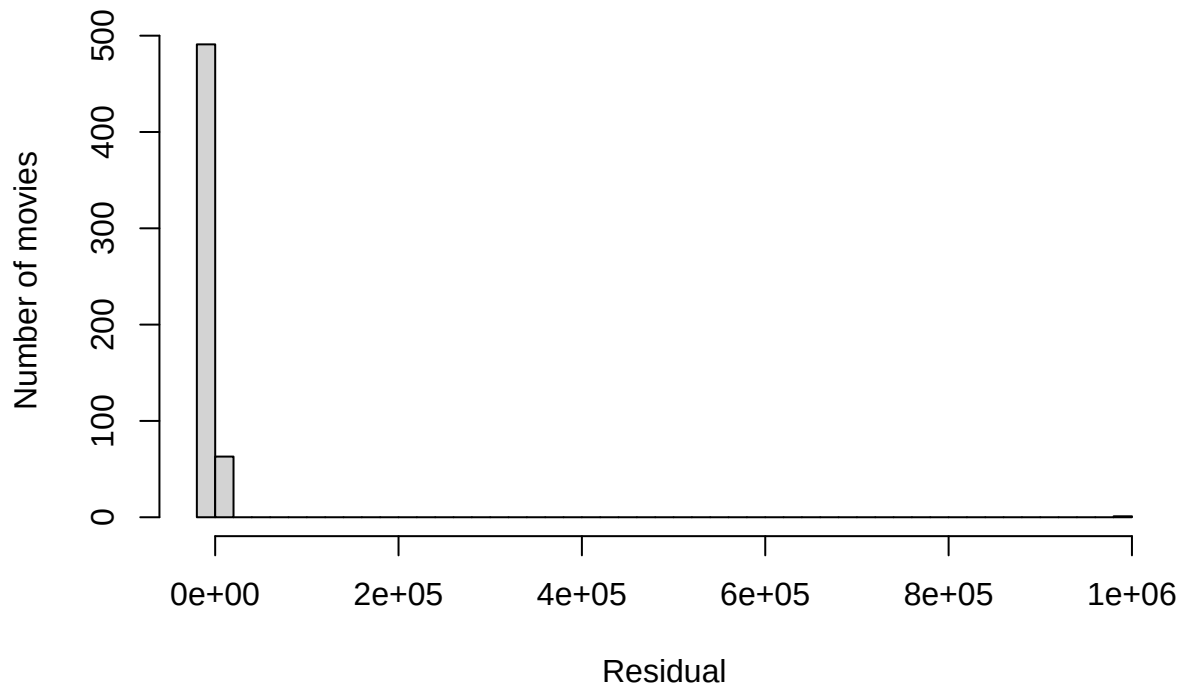
Your Answer:

Chantal's claim is consistent with the data. We cannot reject that PR is unrelated to budget ($p = 0.374$, $R^2 = 0.0014$). Bob's argument that cheaper movies are more profitable is not supported: the slope is negative but not significantly different from zero. Not rejecting H_0 does not prove Chantal right; it only means the data give no evidence against her. My advice is that cutting budgets is not a statistically supported way to raise profitability. The studio should set budgets on other grounds, such as the absolute profit it needs, how much risk it can take, and the genre and audience, rather than expecting cheaper movies to earn a higher PR.

step e

```
#WRITE YOUR CODE HERE  
hist(residuals(model1), breaks = 50, main = "Histogram of residuals (PR ~ budget)",  
      xlab = "Residual", ylab = "Number of movies")
```

Histogram of residuals (PR ~ budget)



```
summary(residuals(model1))
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.     Max.
## -3235   -2866   -2312         0   -1314  996765
```

```
movies[names(which.max(residuals(model1))), c("title", "budget", "revenue", "PR")]
```

```
##           title budget revenue    PR
## 300 Nurse 3-D      10    1e+07 1e+06
```

```
max(rstandard(model1)) # standardized residual of that movie
```

```
## [1] 23.51575
```

Your Answer:

Yes, there is reason to doubt the model. The t-test in question c assumes normally distributed residuals, and these are nowhere near normal.

554 of the 555 residuals sit in the first two bars near 0, between $-3,235$ and $6,359$.

One residual of $996,765$ lies far to the right. It belongs to Nurse 3-D, whose budget is recorded as \$10.

Its standardized residual is 23.5 , while anything above 3 already counts as an outlier.

This single observation pulls the line up. The intercept is 3,235 while a typical PR is about 1.6, so the model overpredicts almost every movie and most residuals are negative (median residual $-2,312$).

The residuals are extremely right-skewed with one huge outlier, so the p-value from c is not reliable.

step f

```
#WRITE YOUR CODE HERE
h <- hatvalues(model1)
n <- length(h)           # 555 observations used
k <- 1                   # one predictor
crit <- 2 * (k + 1) / n   # textbook rule 2(k+1)/n, which is 4/n here
crit
```

```
## [1] 0.007207207
```

```
sum(h > crit)
```

```
## [1] 42
```

```
high_lev <- names(h)[h > crit]
range(movies[high_lev, "budget"])
```

```
## [1] 1.16e+08 2.60e+08
```

```
h[rownames(movies)[movies$title == "Nurse 3-D"]]
```

```
##           300
## 0.003216856
```

Your Answer:

The critical value is $2(k+1)/n = 4/555 = 0.0072$, and 42 observations exceed it. The largest hat value is 0.0485, for Tangled and John Carter. All 42 are big-budget movies, from 116 million to 260 million. Leverage only measures how far a movie's budget lies from the mean budget (\$38.5 million), so these blockbusters sit on the extreme part of the data and can strongly pull the slope. They are real movies, not errors. Still, with 42 potentially influential observations, the slope depends heavily on a small group of expensive films, and that is a reason to doubt the model.

The leverage check does not detect the observation that really distorts the model. Nurse 3-D has a hat value of 0.0032, below 0.0072, because its budget is not unusual on the x-axis. Its problem is its Y-value, as we saw in e).

step g

```
#WRITE YOUR CODE HERE
movies_g <- movies[!(rownames(movies) %in% high_lev), ]
model1g <- lm(PR ~ budget, data = movies_g)
summary(model1g)
```

```
##
## Call:
## lm(formula = PR ~ budget, data = movies_g)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4417  -3731  -2544   -866  995583
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.417e+03  2.926e+03   1.510   0.132
## budget      -8.510e-05  7.561e-05  -1.125   0.261
##
## Residual standard error: 44140 on 511 degrees of freedom
## (152 observations deleted due to missingness)
## Multiple R-squared:  0.002473, Adjusted R-squared:  0.0005205
## F-statistic: 1.267 on 1 and 511 DF, p-value: 0.2609
```

```
# 2) also remove Nurse 3-D: $10 budget is a data error, standardized residual 23.5
model1g2 <- lm(PR ~ budget, data = subset(movies_g, title != "Nurse 3-D"))
summary(model1g2)
```

```
##
## Call:
## lm(formula = PR ~ budget, data = subset(movies_g, title != "Nurse 3-D"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
##  -23.9  -17.3  -11.5   -2.0  4109.5
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.390e+01  1.219e+01   1.961  0.0504 .
## budget      -4.124e-07  3.147e-07  -1.311  0.1906
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 183.5 on 510 degrees of freedom
## (152 observations deleted due to missingness)
## Multiple R-squared:  0.003357, Adjusted R-squared:  0.001403
## F-statistic: 1.718 on 1 and 510 DF, p-value: 0.1906
```

Your Answer:

Without the 42 high-leverage movies ($n = 513$): $PR = 4,417 - 0.0000851 \times \text{Budget}$. For $H_0: \beta_1 = 0$ against $H_1: \beta_1 \neq 0$ at $\alpha = 0.05$, $t = -1.13$ and $p = 0.261$, so we still do not reject H_0 . Also without Nurse 3-D: leverage does not catch the real outlier, so we removed it too. Its 10 dollar budget is probably a data-entry error. The model becomes $PR = 23.9 - 0.00000041 \times \text{Budget}$, which is -0.41 PR per extra 1 million. Now $t = -1.31$ and $p = 0.191$, so it is still not significant, although the coefficients are on a realistic scale.

In every version, budget has no significant effect on PR. the data give no support for Bob's claim that cheaper movies are more profitable.

- a. Make a scatterplot with `vote_average` on the y-axis, and `runtime` on the x-axis. Within the same scatterplot, draw a straight line which summarizes the association between the average vote and the runtime. **[3 points]**
- b. Using the insights from the scatterplot, your linear model, and your data, find an example of a movie that vastly overperformed. That means, find a movie that had much higher average vote than expected, given its runtime. For this movie, report its `vote_average`, its predicted `vote_average`, and its residual. **[3 points]**

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

5 Week 5

- a. Create a plot of the mean PR by year of release. Include 95% confidence intervals around each mean. Do you see any indication of a time trend in the PR? **[4 points]**
- b. Estimate an OLS model which has as dependent variable the PR of a movie, and as independent variable the log of budget, a dummy for each level that `genre2` can take, and the year of release. Show a summary of the resulting model and interpret each coefficient. **[4 points]**

The movie studio that you work is releasing a new movie in May of 2026. It will be a Comedy movie with a budget of 40,000,000. It will have a runtime of 160 minutes.

- c. Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 95% prediction interval. **[6 points]**
- d. Plot the residuals against various independent variables in the model. Do you find any evidence of heteroskedasticity? **[2 points]**
- e. The movie is set in Germany. The director wants to make the movie with only german-speaking actors. The executive producer is strongly against this, as she is afraid that movies that are not english spoken will have a low profit ratio. Estimate a model that can test whether the language spoken impacts the PR of this specific movie. Use an appropriate hypothesis test to test whether the language. What would you advise the movie studio regarding the language to be spoken? **[6 points]**

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step c

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step d

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step e

#WRITE YOUR CODE HERE